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RESEARCH

Efficiency at Maximum Power of a Quantum Stirling Heat Engine

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Abstract
In both the classical and quantum fields, the efficiency at maximum power(EMP) of low-dissipation Carnot heat engines has been extensively studied. However, research on low-dissipation Stirling heat engines has never been done before. We use a two-level system as the working substance to construct a non regenerative quantum Stirling heat engine. In the range of low dissipation, we find that controlling the heat exchange of the working substance during the isochoric process can make the EMP upper bound of the Stirling engine exceed that of the Carnot engine. In addition, we also compare the efficiency of the heat engine in the high-temperature region and the low-temperature region in the case of symmetrical dissipation. The results obtained indicate that the heat engine operating in the high temperature region has an advantage over the low temperature region.

Keywords Stirling cycle · Low-dissipation model · The efficiency at maximum power

1 Introduction

Since the concept of quantum heat engine(QHE) was proposed in 1959 [1], it has undergone extensive attention and vigorous development. As is well known, QHE utilizes quantum systems as working substances to generate work, and the unique quantum properties of the system, such as coherence [2, 3], energy level discretization [4, 5], etc., enable thermal engines to exhibit limits beyond classical thermodynamic settings [6, 7]. Until now, a large number of quantum systems have been used to design different types of quantum heat engines based on different thermodynamic cycles. Including two-level atoms [5, 8–10], spin systems [11, 12], Dirac particles [13], quantum dots [14–16], and ion traps [17]. These quantum heat engines provide a new perspective and direction for verifying quantum thermodynamic experiments, understanding the second law of thermodynamics, describing non-equilibrium thermalization and fluctuations of quantum systems, and other thermodynamically related theories [2–17].

In fact, the development of heat engines has gone through a long stage. In its early days, most research focused on equilibrium thermodynamics, where the second law of thermo-

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dynamics stated that a heat engine operating between hot bath T_h and cold bath T_c would be limited by Carnot efficiency $\eta_c = 1 - \frac{T_c}{T_h}$ $(T_c < T_h)$ [18]. This upper bound is universal and independent of the characteristics of heat engines, including quantum heat engines [19]. Although quasi-static heat engines have the highest efficiency, their output power completely disappears because the time required to complete the entire cycle is infinitely long. From both theoretical and practical perspectives, we should accelerate the cycle of the heat engine to obtain limited power, so it is worthwhile to study the efficiency at maximum power (EMP) within the framework of finite time thermodynamics [5, 8, 9, 20–26]. One of the most typical works is the internally reversible model proposed by Curzon and Ahlborn(CA), who found that the EMP of a heat engine can be represented as $\eta_{CA} = 1 - \sqrt{\frac{T_c}{T_h}}$. This is the famous CA efficiency [23]. Since then, people have conducted in-depth research on the universality of CA efficiency and gradually transitioned to quantum systems. It can be known that finite time thermodynamics is a very useful tool for optimizing heat engines, such as Carnot engines [24–26], Otto engines [9, 11], and Stirling engines [27–30], among others. However, in previous extensive research, the optimization of EMP has mainly focused on the Carnot and Otto engines, with relatively little attention paid to the Stirling engine. And even if there were, most of the research focused on quasi-static Stirling heat engines [27–29, 31–35]. There are only a handful of heat engine studies with real power output. Recently, based on an opto-mechanical system platform, resonators that interact with feedback-controlled optical cavities by radiant pressure have achieved efficient thermodynamic conversion using the Stirling cycle [36]. Using a two-level system(TLS) as the working medium, Raja et al. studied the thermodynamic properties of a finite-time non-regenerative quantum Stirling cycle [37]. Similarly, in a finite time, the efficiency of the Stirling heat engine operating at low temperatures can be close to the Carnot efficiency by changing the energy level structure of the particles by using a barrier that changes rapidly with time. Das et al. [30].These works have attracted more attention.

Like the Carnot cycle, the Stirling cycle also includes isothermal processes. However, dealing with isothermal processes in a finite time is more complex and requires you to obtain real-time kinetics of the working substance in a finite time. Thus, we use a discrete isothermal process (DIP), which consists of a series of isochoric and adiabatic processes. In a large number of previous studies, it has been considered as a good alternative to isothermal processes [5, 32, 38]. Since there is no time consumption in the adiabatic process in DIP, we use the Markov main equation to simulate the dynamic changes of the working substance when it is coupled with the heat bath in the isochoric process, and then obtain the dynamic evolution of the working substance [11]. In fact, the performance of low dissipation Carnot engines has been widely discussed in the past decade. These researches include the equivalence of the model in a linear response system [39], the power-efficiency trade-off optimization and constraints [40], and the effects of adiabatic process dissipation [41] and so on. Of course, the main direction is to find a common trade-off between power and efficiency for the heat engine, and these efforts provide important guidance for the optimization of the heat engine. [40, 42–46]. In previous studies, the $1/\tau$-scaling of low-dissipation models based on two-level systems has been rigorously proved and verified both theoretically and experimentally [40, 47, 48]. Moreover, this mechanism is valid in the long-time regime. Therefore, in this paper, we also apply the $1/\tau$-scaling to the Stirling cycle, and it should be emphasized that the low dissipation Stirling cycle we constructed is reasonable, as each cycle process has its corresponding physical mechanism.

The paper is organized as follows:In Section 2, we introduce the master equation and then construct the non-regenerative Stirling heat engine in Section 3. Also in Section 3, on the basis of the generalized maximum power, we carefully discuss the efficiency boundary range of the low-dissipation Stirling heat engine under extreme dissipation conditions, and obtain some interesting results. Theoretical calculations and numerical simulations of heat engines are discussed in Section 4. Finally, Section 5 is the conclusion and outlook.

2 Evolution of a Two-Level System in a thermal reservoir

We first consider a working fluid consisting of many noninteracting spin-$1/2$ systems. The magnetic field $\mathbf{B}$ can be used as an external control parameter of the system, with its direction remains unchanged and along the positive-z axis. For a spin system with a magnetic moment $\mathbf{M}$ trapped in a magnetic field $\mathbf{B}$, the system Hamiltonian is given by $\hat{H}(t) = -\mathbf{M} \cdot \mathbf{B} = 2\mu_B \mathbf{S} \cdot \mathbf{B} = 2\mu_B B_z(t) S_z$, where μ_B is the Bohr magnetron, $\mathbf{S}$ is the spin angular momentum. We define $\omega(t) = 2\mu_B B_z(t)$, the average spin polarization S can be given by $S = \langle S_z \rangle = -\frac{1}{2}\tanh\left(\frac{\beta\omega}{2}\right)$, where $\beta = 1/T$ is the inverse temperature of the system, here, we set $k_B = 1$. The internal energy of the spin systems is defined as the expectation value of the Hamiltonian $(\hbar \equiv 1)$:

$$U = \langle H \rangle = \omega S = -\frac{1}{2}\omega \tanh\left(\frac{\beta\omega}{2}\right). \tag{1}$$

For open quantum systems, the dynamical evolution in contact with a heat bath is extremely complex. However, the dynamics of open quantum systems far from thermal equilibrium can be modeled in the form of semigroup formalism [12, 24]. Here, we use the Heisenberg picture to consider the time evolution of spin systems.

First, the state of the ensemble of systems is described by the density operator $\hat{\rho}$,the expected value of an operator at time t is given by

$$\left\langle \hat{X}(t) \right\rangle = Tr\left[\hat{\rho}\hat{X}(t)\right]. \tag{2}$$

It should be noted that the expected value of $\left\langle \hat{X} \right\rangle$ is also simply denoted by $\hat{X}$.Therefore,the motion equation of an operator in the Heisenberg picture can be written as [11, 12]

$$\frac{d}{dt}\hat{X}(t) = i\left[\hat{H}(t), \hat{X}(t)\right] + \mathcal{L}_D^*\left[\hat{X}(t)\right] + \frac{\partial}{\partial t}\hat{X}(t) \tag{3}$$

where $\mathcal{L}_D^* = \sum_\alpha k_\alpha \left(\hat{V}_\alpha^\dagger\left[\hat{X}, \hat{V}_\alpha\right] + \left[\hat{V}_\alpha^\dagger, \hat{X}\right]\hat{V}_\alpha\right)$ is the superoperator that represents the nonunitary part of the evolution caused by dissipative interactions between the system and the heat reservoir, $\partial\hat{X}(t)/\partial t$ is the explicit time dependence of the operator. Here $\hat{V}_\alpha^\dagger$ and $\hat{V}_\alpha$ are operators in the Hilbert space of the system and are Hermitian conjugates and k_α are known as transition rates.In a spin system, the specific form of $\mathcal{L}_D^*$ can be represented as

$$\mathcal{L}_D^*\left[\hat{X}(t)\right] = k_\uparrow\left(\hat{S}\left[\hat{X}, \hat{S}^\dagger\right] + \left[\hat{S}, \hat{X}\right]\hat{S}^\dagger\right) + k_\downarrow\left(\hat{S}^\dagger\left[\hat{X}, \hat{S}\right] + \left[\hat{S}^\dagger, \hat{X}\right]\hat{S}\right), \tag{4}$$

the operators $\hat{V}_\alpha^\dagger$ and $\hat{V}_\alpha$ become the spin creation $\hat{S}^\dagger = \hat{S}_x + i\hat{S}_y$ and annihilation operators $\hat{S} = \hat{S}_x - i\hat{S}_y$. We set $\hat{X} = S_z$ and substitute it in (3),the instantaneous polarization relaxation equation can be written as

$$\frac{dS}{dt} = \langle \mathcal{L}_D^*(S_z) \rangle = -2\left(k_\uparrow + k_\downarrow\right)S - \left(k_\downarrow - k_\uparrow\right). \tag{5}$$

Here, we require that ω be constant, and $k_\uparrow$ and $k_\downarrow$ are also constant. The general solution of (5) is straightforward

$$S(t) = S^{eq} + \left[S(0) - S^{eq}\right] e^{-\gamma t} \tag{6}$$

where $\gamma = k_\uparrow + k_\downarrow$is the heat conductivity for the spin system and $S^{eq} = -\frac{1}{2}\frac{k_\downarrow - k_\uparrow}{k_\uparrow + k_\downarrow}$ is the asymptotic population, which corresponds to the value at thermal equilibrium state. In addition, to ensure that the system asymptotically evolves to the correct equilibrium state in a specific way, we also require that $k_\uparrow$ and $k_\downarrow$ must satisfy the relationship [12]

$$\frac{k_\downarrow}{k_\uparrow} = e^{\beta\omega} \tag{7}$$

In fact, the dynamic information related to this relationship must be based on more detailed information, which limits the model of the heat bath and its coupling with the spin system.

3 Low-dissipation Stirling Heat Engine Model

The classical Stirling cycle consists of two isothermal stages and two isochoric thermalizations. Traditionally, the cycle typically requires two additional stages to replace the lost heat, which includes the interaction of the working substance with a so-called regenerator. As a result, this requires the regenerator to have a very high heat capacity in order to absorb heat during the cooling isochoric stroke and transfer the heat back to the working substance when heating the isochor, thus increasing overall efficiency and minimizing waste heat. In the Stirling cycle model in this paper, we do not consider the regeneration device, and the working substance directly interacts with the hot bath to transfer heat.

Now, we have developed an easily operable two-level heat engine using a spin system as the working substance, aiming for applications in optimizing finite-time Stirling heat engines. The Stirling cycle consists of four stages, two isothermal stages and two isochoric stages, which is given in Fig. 1a. In the figure, ω_2 and ω_1 are the initial and final energy spacing of the working substance during the isothermal process, respectively. We also require the system to

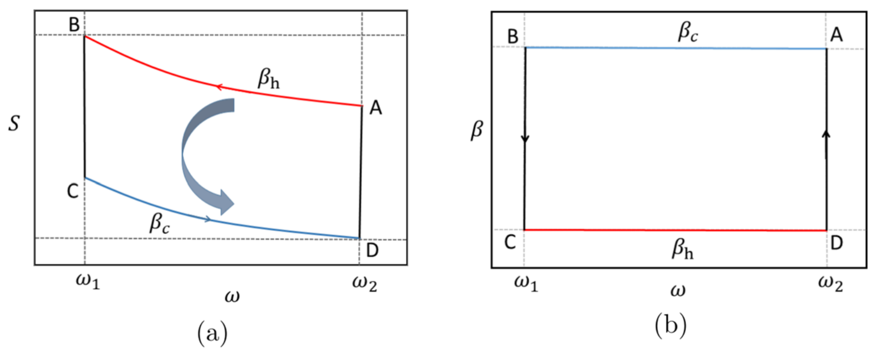

Fig. 1 (Color online) (a) Schematic diagram of a Stirling cycle in the (ω, S). The level spacing and the average spin polarization of the system are represented by ω and S, respectively. The red and blue solid curves represent the isothermal processes at high and low temperatures for finite time, respectively. The black (vertical) solid line represents the isochoric process. (b) Schematic diagram of a Stirling cycle in the (ω, β) plane. In the isothermal process, the inverse temperature remains constant, and the inverse temperature of the isochoric process changes. At each endpoint, the TLS is in equilibrium with the heat bath

be in equilibrium at each endpoint for ease of control. In the Fig. 1b, during the isothermal process, the reverse temperature $\beta_h(\beta_c)$ of the working substance remains constant, and in the isochoric process, its inverse temperature changes. In order to ensure that the classical isothermal process is operated within a finite duration, we chose the discrete isothermal process as an alternative. As shown in Fig. 2a, the DIP includes a series of N alternating short adiabatic processes and isochoric processes. Every quantum adiabatic process is operated by sudden quenching and has no time cost. In general, a quantum adiabatic process requires a sufficiently long time so that the instantaneous eigen-states of the Hamiltonian $H(t)$ of the system remains unchanged during the process [11]. But in some cases we can tune the energy spacing of the system fast enough so that the population of the system does not change with time. In other words, the rapid changes in control parameters correspond to $\dot{\omega} \to \infty$, where ω changes very fast on the time scale of the mean population thermal relaxation [5, 8]. Due to the absence of level crossing in spin-1/2 systems, the adiabatic process in this paper is also applicable to such situations.

It's interesting to note that the benefit of discrete isothermal processes is that the work and heat exchanged by the system with its surroundings are separated. In the quantum isochoric process, no work is generated, it only has heat exchange, in fact the work only occurs in the quantum adiabatic process. Furthermore, we also choose the maximum step time $\tau_\alpha^{\max}$ to replace the time τ_j $(j = 1, 2, ..., N)$ for each step of the isochoric process. According to (6), we can take the relaxation time $\frac{2}{\gamma}$ for each step of the isochoric process. Then the average spin polarization $S(t)$ asymptotically evolves into S^{eq}. The problem now is that the relaxation time of each step is not the same, which leads to the fact that the total time of the isothermal process cannot be determined. For this purpose, we chose the maximum step length $\tau_j^{\max}$ as the relaxation time for each step. Using (7), we can get $\tau_j^{\max} = \frac{2}{k_\uparrow(1+e^{\beta\omega_2})}$. Therefore, the total time of the isothermal process can be written as $t_\alpha = N\tau_\alpha^{\max} = \frac{2N}{k_\uparrow(1+e^{\beta\omega_2})}$. In the past, there were many ways to tune the energy level, and in order to obtain the minimum irreversible entropy of the isothermal process, it was necessary to use the external method of linear control, that is, the energy level was changed at equal intervals [5]. For the operation of the heat engine in this paper, we also adopt this method. The cycles with different time intervals are illustrated in Fig. 2b. We can see that the energy level spacing of the system

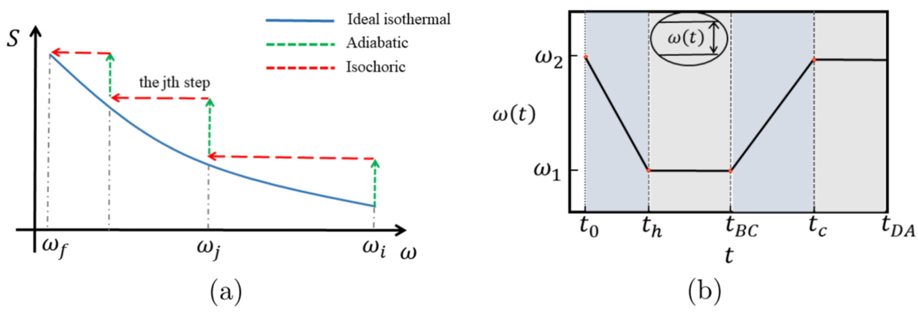

Fig. 2 (Color online) (a) Schematic diagram of a discrete isothermal process (DIP). The solid blue line represents the ideal isothermal process, which follows Boltzmann distribution. The green vertical dashed line and the red horizontal dashed line represent the isochoric process and the adiabatic process, respectively. (b) Schematic diagram of the change of energy level spacing $\omega(t)$ with time t. The corresponding stage intervals of the isothermal process are $[t_0, t_h]$ and $[t_{BC}, t_c]$, respectively. $[t_h, t_{BC}]$ and $[t_c, t_{DA}]$ are the stage intervals of the isochoric process. The black solid curve shows the change in the energy spacing

varies linearly and is represented by a black solid curve. Each stage corresponds to a time interval, for a total of four intervals.

Our cycle begins in an equilibrium state at temperature β_h and then goes through the following four stages:

(i) **Isothermal compression: $A\ (\omega_2, T_h) \rightarrow B\ (\omega_1, T_h)$.** Initially, the working substance is in equilibrium with a hot bath with temperature T_h. Subsequently, the working substance comes into contact with the hot bath within time interval t_h. The level spacing of the TLS is reduced from ω_2 to ω_1. Throughout the stage, the external control parameters are continuously reduced. Then, according to the mechanism in which the dissipation coefficient Σ is inversely proportional to time [8, 26, 47], the heat exchanged by the heat engine with the heat bath during isothermal compression is

$$Q_h = T_h \Delta S_h - T_h \frac{\Sigma_h}{t_h}, \tag{8}$$

where ΔS_h is the change in entropy of the working substance during isothermal compression. It should be noted that in previous studies on two-level heat engines, ΔS_α and $\Sigma_\alpha (\alpha = h, c)$ are usually related to ω and γ [5, 8, 40]. However, considering the convenience of optimization, we have not written their specific forms here, but have replaced them with generalized expressions.

(ii) **Isochoric compression: $B\ (\omega_1, T_h) \rightarrow C\ (\omega_1, T_c)$.** In the second stage, with duration t_{BC}, the TLS is disconnected from the hot bath and is connected with the cold bath at temperature T_c with fixed frequency ω_1. The system does no work and releases heat to the environment. At the end of the stage system reaches thermal equilibrium with the cold bath at the temperature T_c. We assume that the heat Q_{BC} exchanged by the system with the cold bath is proportional to the heat exchanged by the isothermal expansion process, which is given by

$$Q_{BC} = \phi Q_c = \phi \left(T_c \Delta S_c - T_c \frac{\Sigma_c}{t_c} \right). \tag{9}$$

where the scale factor $0 < \phi$.

(iii) **Isothermal expansion: $C\ (\omega_1, T_c) \rightarrow D\ (\omega_2, T_c)$.** In the third stage, at temperature T_c, the working substance remains in contact with the cold bath within the time interval t_c. The level spacing of the TLS increases from ω_1 back to ω_2 while it is still coupled to the cold bath. The total heat exchanged Q_{CD} between the system and the hot bath can be written as

$$Q_c = T_c \Delta S_c - T_c \frac{\Sigma_c}{t_c}. \tag{10}$$

where ΔS_c is the change in entropy of the working substance during isothermal expansion.

(iv) **Isochoric expansion: $D\ (\omega_2, T_c) \rightarrow A\ (\omega_2, T_h)$.** Finally, with duration t_{DA}, at a constant frequency ω_2, the TLS is disconnected from the cold bath and is connected with the hot bath at an inverse temperature β_h. Here, we assume that Q_{DA} can be written as

$$Q_{DA} = \kappa Q_h = \kappa \left(T_h \Delta S_h - T_h \frac{\Sigma_h}{t_h} \right). \tag{11}$$

where the scale factor $0 < \kappa$. After the system completes the entire cycle, the total change in entropy of the working substance is zero. It should be noted that due to the isochoric process throughout the entire cycle, the working substance comes into direct contact with the heat bath, which may result in situations where ϕ and κ are greater than 1. We also assume that the time of two isochoric processes is proportional to the time of two isothermal processes,

$$t_{BC} + t_{DA} = \chi \left(t_h + t_c\right) \tag{12}$$

where $\chi > 0$ is a constant. During the total time period $t_{cycle} = t_h + t_c + \chi \left(t_h + t_c\right)$, the total work done by the heat engine throughout the entire cycle is $-W = Q_h + Q_{BC} + Q_c + Q_{DA}$. We have adopted the usual convention here that the heat and work absorbed by the system is positive. When substituting $-W$ into $P = \frac{-W}{t_{cycle}}$, the power produced throughout the Stirling cycle is given by

$$P = \frac{-W}{\left(1+\chi\right)\left(t_h + t_c\right)} = \frac{\left(\kappa+1\right)\left(T_h \Delta S_h - \frac{T_h \Sigma_h}{t_h}\right) + \left(\phi+1\right)\left(T_c \Delta S_c - \frac{T_c \Sigma_c}{t_c}\right)}{\left(1+\chi\right)\left(t_h + t_c\right)}. \tag{13}$$

The efficiency of the heat engine is then given by

$$\eta = \frac{Q_h + Q_{BC} + Q_c + Q_{DA}}{Q_h + Q_{DA}} = 1 + \frac{Q_c + Q_{BC}}{Q_h + Q_{DA}} \tag{14}$$

where $Q_{ins} = Q_h + Q_{DA}$, $Q_{out} = Q_c + Q_{BC}$. Next, we substitute the concrete forms of Q_{ins} and Q_{out} into the (14), and we get

$$\eta = 1 + \frac{\left(\phi+1\right)\left(T_c \Delta S_c - \frac{T_c \Sigma_c}{t_c}\right)}{\left(\kappa+1\right)\left(T_h \Delta S_h - \frac{T_h \Sigma_h}{t_h}\right)} \tag{15}$$

We use t_h and t_c as optimization parameters, maximizing P with respect to them, i.e., setting $\frac{\partial P}{\partial t_h} = 0$ and $\frac{\partial P}{\partial t_c} = 0$. First of all, we find that the ratio of t_c^* and t_h^* is

$$\frac{t_c^*}{t_h^*} = \sqrt{\frac{\left(1+\phi\right)}{\left(1+\kappa\right)}}\sqrt{\frac{T_c \Sigma_c}{T_h \Sigma_h}}. \tag{16}$$

Then, we obtain the optimal operating time, respectively

$$t_h^* = \frac{2\left(1+\kappa\right) T_h \Sigma_h}{\left(1+\kappa\right) T_h \Delta S_h + \left(1+\phi\right) T_c \Delta S_c}\left(1 + \sqrt{\frac{\left(1+\phi\right)}{\left(1+\kappa\right)}}\sqrt{\frac{T_c \Sigma_c}{T_h \Sigma_h}}\right) \tag{17}$$

and

$$t_c^* = \frac{2\left(1+\phi\right) T_c \Sigma_c}{\left(1+\kappa\right) T_h \Delta S_h + \left(1+\phi\right) T_c \Delta S_c}\left(1 + \sqrt{\frac{\left(1+\kappa\right)}{\left(1+\phi\right)}}\sqrt{\frac{T_h \Sigma_h}{T_c \Sigma_c}}\right). \tag{18}$$

Since the entropy change of the two isothermal processes is not equal, we set $\Delta S_h = \alpha \Delta S_c, \alpha$ is a negative number, i.e. $\alpha < 0$. By using the above equations and substituting two optimization times, the efficiency at maximum power η_S can be obtained and is given by

$$\eta_S = \frac{\left(1+\kappa\right) + \frac{1}{\alpha}\left(1+\phi\right)\left(1-\eta_c\right)\left[\left(1+\kappa\right) + \sqrt{\left(1+\kappa\right)\left(1+\phi\right)}\sqrt{\frac{T_c \Sigma_c}{T_h \Sigma_h}}\right]}{\left[\left(1+\kappa\right) + \sqrt{\left(1+\kappa\right)\left(1+\phi\right)}\sqrt{\frac{T_c \Sigma_c}{T_h \Sigma_h}}\right]^2 + \left(1+\kappa\right)\left(1+\phi\right)\frac{T_c}{T_h}\left(-\frac{1}{\alpha} - \frac{\Sigma_c}{\Sigma_h}\right)} \tag{19}$$

and the power generated during the Stirling cycle can be written as

$$P_{\max}^{S}=\frac{\left[\alpha\left(1+\kappa\right)+\left(1+\phi\right)\left(1-\eta_{c}\right)\right]^{2}T_{h}^{2}\Delta S_{c}^{2}}{4\left(1+\chi\right)\left[\sqrt{\left(1+\kappa\right)T_{h}\Sigma_{h}}+\sqrt{\left(1+\phi\right)T_{c}\Sigma_{c}}\right]^{2}} \tag{20}$$

Interestingly, in the above series of results, we can see that χ only has a significant impact on power. This means that we can ignore the influence of χ in optimizing the efficiency of the heat engine. In the asymmetric dissipation limits, the generalized extreme bounds of the efficiency at maximum power are obtained

$$\frac{1+\frac{L}{\alpha}\left(1-\eta_{c}\right)}{1+\sqrt{-\frac{L}{\alpha}}}=\eta_{S}^{-}\leq\eta_{S}\leq\eta_{S}^{+}=\frac{1+\frac{L}{\alpha}\left(1-\eta_{c}\right)}{1-\frac{L}{\alpha}\left(1-\eta_{c}\right)}, \tag{21}$$

where $L=\frac{1+\phi}{1+\kappa}$. The upper bound η_S^+, which is reached in the completely asymmetric limit $\frac{\Sigma_c}{\Sigma_h}\rightarrow 0$, and $\frac{\Sigma_c}{\Sigma_h}\rightarrow\infty$ corresponds to the lower boundary η_S^-. From (21), we can get that the values of L and α affect the efficiency boundary of the low-dissipation Stirling cycle. It is also important to note that the values of the two parameters are not arbitrary, and they are also subject to boundary conditions. Thus, for α, when the system completes the entire cycle, its overall entropy becomes zero, which means that $\Delta S_h+\Delta S_c+\Delta S_{CB}+\Delta S_{AD}=0$. Then we have

$$\alpha=-\left(1+\frac{\Delta S_{CB}+\Delta S_{AD}}{\Delta S_{c}}\right) \tag{22}$$

and for L, constrained by boundary condition: $0<\eta_S^-<\eta_S^+<\eta_c$, we get an inequality

$$\frac{-\alpha}{1+\eta_{c}}=L^{-}<L<L^{+}=\frac{-\alpha}{1-\eta_{c}}. \tag{23}$$

Now, we wonder if the upper bound of the efficiency of (21) will be greater than $\frac{\eta_c}{2-\eta_c}$? The results are there. Using the relation $\eta_S^+\geq\frac{\eta_c}{2-\eta_c}$, we can obtain $L\leq-\alpha$. What we do know is that for low dissipation Carnot heat engines, as long as we let $L=1, \kappa=0,\ \phi=0,\ \alpha=-1$, the efficiency bounds of (21) return to the bounds of the low-dissipation Carnot cycle, namely,

$$\frac{\eta_{c}}{2}=\eta_{C}^{-}\leq\eta_{C}\leq\eta_{C}^{+}=\frac{\eta_{c}}{2-\eta_{c}}. \tag{24}$$

In this case, $\frac{L}{\alpha}=-1$. This shows that (21) has a certain universality. Therefore, as long as $L<-\alpha$ and the parameters are limited by certain boundary conditions, the efficiency of this type of Stirling heat engine will exceed the upper bound of the low-dissipation Carnot heat engine. It is an important result obtained in this paper.

Next, we will discuss some of the properties of the low-dissipation Stirling heat engine in Section (4) below, based on the platform of two-level systems.

4 Theoretical Calculations and Numerical Simulations of Finite-time Stirling Engines

As the counterpart of classical working substance, quantum system has different working efficiency and power compared with classical heat engine because of its unique properties [5, 6, 8, 35]. Therefore, we use a two-level system as the working substance to achieve a low-dissipation Stirling cycle and understand some of the performance of the heat engine through theoretical and numerical calculations. In Fig. 1a, the TLS is in equilibrium at each endpoint, the equilibrium states corresponding to the two levels can be expressed as

$$P_g = \frac{e^{\beta_\alpha \omega}}{1 + e^{\beta_\alpha \omega}} \quad P_e = \frac{1}{1 + e^{\beta_\alpha \omega}} \tag{25}$$

where $\alpha = (h, c)$. The Shannon entropy of the system can be written as

$$S = -\sum p_i \ln p_i (k_B = 1) \tag{26}$$

where P_i is the ith eigenstate of the spin system. Substituting (25) into (26), the representation of the entropy function is given by

$$S(\beta\omega) = \ln\left(1 + e^{\beta\omega}\right) - \frac{\beta\omega e^{\beta\omega}}{1 + e^{\beta\omega}}. \tag{27}$$

Next, according to the constraints of parameter α, we calculate the entropy changes corresponding to the four stages of the cyclic process.

First of all, we consider the case in the high temperature limit $\beta\omega \ll 1$. We take the Taylor expansion of (27) and keep it to the second order term

$$S\left(\beta\omega\right) = \left[\ln\left(2\right)\right]_{\beta\omega=0} - \left[\frac{1}{8}\right]_{\beta\omega=0} \left(\beta\omega\right)^2 + O\left[\left(\beta\omega\right)^3\right]. \tag{28}$$

Hence, the entropy change ΔS_α in isothermal processes can be analytically written as

$$\Delta S_h = \frac{\beta_h^2}{8}\left(\omega_2^2 - \omega_1^2\right), \ \Delta S_c = -\frac{\beta_c^2}{8}\left(\omega_2^2 - \omega_1^2\right), \tag{29}$$

and the entropy change of the two isochoric processes can be written as

$$\Delta S_{CB} = \frac{\omega_1^2}{8}\left(\beta_h^2 - \beta_c^2\right), \ \Delta S_{AD} = -\frac{\omega_2^2}{8}\left(\beta_h^2 - \beta_c^2\right), \tag{30}$$

By substituting (30) and (29) into (22) , we can obtain

$$\alpha_{High} = -\frac{\beta_h^2}{\beta_c^2} = -\left(1 - \eta_c\right)^2. \tag{31}$$

Substituting (31) into (21) , the extreme boundary of efficiency of the Stirling cycle at the high temperature limit can be expressed as

$$\frac{(1-\eta_c) - L}{(1-\eta_c) + \sqrt{L}} \leq \eta_S \leq \frac{(1-\eta_c) - L}{(1-\eta_c) + L} \tag{32}$$

In this way, the (23) becomes

$$\frac{(1-\eta_c)^2}{(1+\eta_c)} = L^-_{high} < L < L^+_{high} = (1-\eta_c). \tag{33}$$

So we conclude that at the high temperature limit, the efficiency η_S boundary of the Stirling cycle remains variable due to the influence of the coefficient L, but the range of L values is only related to η_C and independent of the energy level spacing ω. Next, we will consider the low-temperature case $\beta\omega \gg 1$. The entropy of a two-level system in equilibrium can be written as

$$S\left(\beta\omega\right) = \beta\omega e^{-\beta\omega} \tag{34}$$

Similarly, the entropy change ΔS_α in isothermal processes can be written as

$$\Delta S_h = \beta_h \omega_1 e^{-\beta_h \omega_1} - \beta_h \omega_2 e^{-\beta_h \omega_2}, \quad \Delta S_c = \beta_c \omega_2 e^{-\beta_c \omega_2} - \beta_c \omega_1 e^{-\beta_c \omega_1}, \tag{35}$$

and for the entropy change in the isochoric process, we have

$$\Delta S_{CB} = \beta_c \omega_1 e^{-\beta_c \omega_1} - \beta_h \omega_1 e^{\beta_h \omega_1}, \quad \Delta S_{AD} = \beta_h \omega_2 e^{-\beta_h \omega_2} - \beta_c \omega_2 e^{-\beta_c \omega_2} \tag{36}$$

Substituting (35) and (36) into (22) , we can end up with

$$\alpha_{Low} = \frac{\beta_h \omega_1 e^{-\beta_h \omega_1} - \beta_h \omega_2 e^{-\beta_h \omega_2}}{\beta_c \omega_2 e^{-\beta_c \omega_2} - \beta_c \omega_1 e^{-\beta_c \omega_1}}. \tag{37}$$

Therefore, the efficiency boundary of the Stirling heat engine at low temperature can be expressed as

$$\frac{1 + \frac{L}{\alpha_{Low}} (1 - \eta_c)}{1 + \sqrt{-\frac{L}{\alpha_{Low}}}} =\leq \eta_S \leq \frac{1 + \frac{L}{\alpha_{Low}} (1 - \eta_c)}{1 - \frac{L}{\alpha_{Low}} (1 - \eta_c)}. \tag{38}$$

Unlike in the case of high temperatures, the (23) becomes

$$\left(\frac{1 - \eta_c}{1 + \eta_c}\right) \frac{e^{\beta_c(\omega_1+\omega_2)} \left[e^{-\beta_h \omega_2} - \frac{\omega_1}{\omega_2} e^{-\beta_h \omega_1}\right]}{e^{\beta_c \omega_1} - \frac{\omega_1}{\omega_2} e^{\beta_c \omega_2}} =< L^-_{low} < L < L^+_{low} = \frac{e^{\beta_c(\omega_1+\omega_2)} \left[e^{-\beta_h \omega_2} - \frac{\omega_1}{\omega_2} e^{-\beta_h \omega_1}\right]}{e^{\beta_c \omega_1} - \frac{\omega_1}{\omega_2} e^{\beta_c \omega_2}} \tag{39}$$

The above results indicate that the boundary between efficiency η_S and coefficient L at low temperatures is not only related to η_C, but also closely related to the level spacing ω. We show in Fig. 3 the coefficient L as a function of η_C for both high and low temperature cases. Here, we choose the initial and final energy level spacing of the two-level system, $\omega_1 = 0.2$ and $\omega_2 = 0.4$, respectively. The Fig. 3 shows that, in the low-temperature region, L mainly remains greater than 1, while in the case where η_C is small, L can be less than 1. However, overall $\phi > \kappa$. In high-temperature areas, $L < 1$, hence, $\phi < \kappa$. Meanwhile, the orange part of Fig. 3 also indicates that if we want to exceed the efficiency upper bound of a low dissipation Carnot heat engine, in the high temperature region, κ must always be greater than ϕ, while in the low temperature region, κ is usually less than ϕ, except for low Carnot efficiency.

Now, we discuss the change in efficiency of the heat engine at maximum power with Carnot efficiency throughout the cycle in the case of symmetric dissipation ($\Sigma_c = \Sigma_h$).

In the previous discussion, we found that when $\frac{L}{\alpha} = -1$, the efficiency limit of the Stirling heat engine regresses to that of the Carnot heat engine, which is an important condition. Therefore, we assume that when the heat engine is in the case of symmetric dissipation, it also satisfies this condition. In the high-temperature region, $\alpha_{High} = -(1 - \eta_c)^2$, at this time, the efficiency of the heat engine at maximum power is

$$\eta^*_{S-High} = \frac{\eta_c}{1 + (1 - \eta_c)\sqrt{(1 - \eta_c)} + \frac{\eta_c(1-\eta_c)(2-\eta_c)}{1+(1-\eta_c)\sqrt{(1-\eta_c)}}}. \tag{40}$$

In this region, $\phi < \kappa$ always holds. Similarly, in the field of low temperatures, the efficiency of the heat engine at maximum power can be expressed as

$$\eta^*_{S-Low} = \frac{\eta_c}{1 + \sqrt{-\alpha_{Low}(1 - \eta_c)} + \frac{(1-\eta_c)(1+\alpha_{Low})}{1+\sqrt{-\alpha_{Low}(1-\eta_c)}}} \tag{41}$$

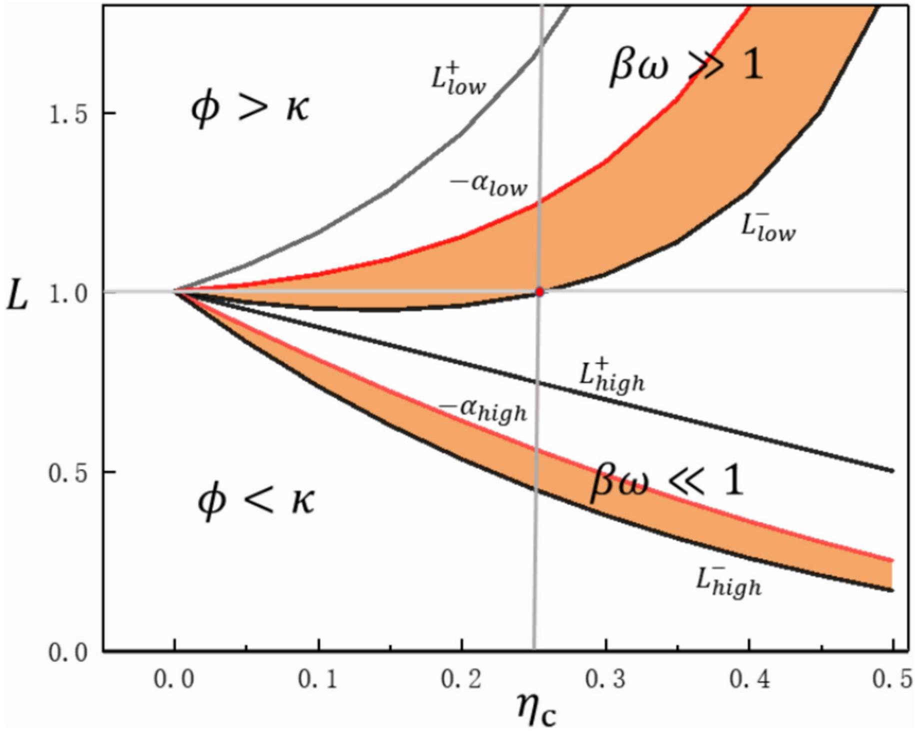

Fig. 3 (color online) The coefficient L as a function of η_C. $L^{\pm}_{low}$ and $L^{\pm}_{high}$ correspond to the low temperature region($\beta\omega \gg 1$) and the high temperature region($\beta\omega \ll 1$), respectively. In the figure, we use $L = 1$ as the reference line. Thus, when $L > 1, \phi > \kappa$; when $L < 1, \phi < \kappa$. The red solid curve corresponds to the change of $-\alpha$ with η_C in both cases. In the orange area, L satisfies $L^{-} < L \leq -\alpha$. In this region, the efficiency of the Stirling heat engine at maximum power is greater than or equal to $\frac{\eta_c}{2-\eta_c}$

where $\alpha_{Low} = \frac{\beta_h\omega_1 e^{-\beta_h\omega_1} - \beta_h\omega_2 e^{-\beta_h\omega_2}}{\beta_c\omega_2 e^{-\beta_c\omega_2} - \beta_c\omega_1 e^{-\beta_c\omega_1}}$. Through the above two equations, we can see that the efficiency of the two extreme cases is closely related to η_C. We plot the efficiency of the maximum power of the Stirling heat engine as a function of η_C in Fig. 4. In the simulation, the parameters are set as $\omega_1 = 0.2$, $\omega_2 = 0.4$, they remain the same as the previous settings. And in high-temperature areas, $\beta_h = 0.01$ is fixed, in low-temperature areas, $\beta_h = 10$ is also fixed. Their corresponding β_c constantly changes with η_C. As can be seen from the figure, in the high temperature region, in the small Carnot efficiency range, the symmetrically dissipated Stirling heat engine efficiency is close to the Curzon Ahlborn efficiency, while in the large η_C range, η^{*}_{S-High} exceeds it. However, in the low temperature region, the situation is different, we can see that in the small η_C range, η^{*}_{S-Low} is also close to the CA efficiency , but in the large η_C range, the efficiency of the heat engine is always less than the efficiency value of CA and is infinitely close to the value of 0.5. This indicates that in the small Carnot range, the efficiency of the two extreme cases at maximum power is similar, while in the large Carnot range, the heat engine operating in the high-temperature region has an advantage over the low-temperature region.

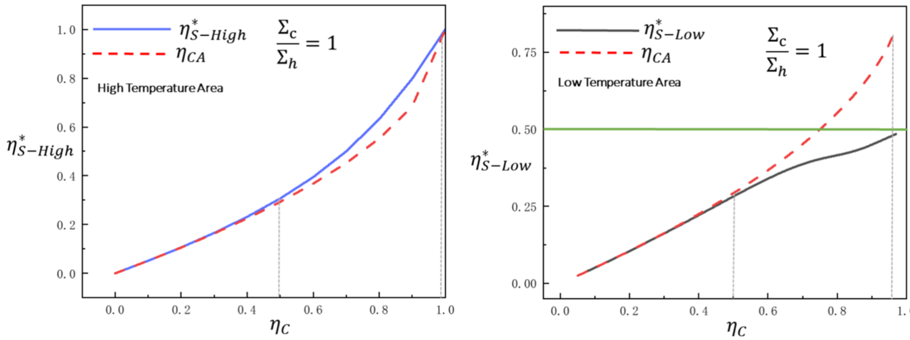

Fig. 4 (Color online) Efficiency at maximum power as a function of η_C. In the case of symmetric dissipation($\frac{\Sigma_c}{\Sigma_h} = 1$), the Curzon-Ahlborn efficiency is plotted with a red dash line. The numerically obtained η^*_{S-Low} and η^*_{S-High} are plotted with the blue solid line and black solid line, respectively. In this example, we choose $\omega_1 = 0.2$, $\omega_2 = 0.4$, and the inverse temperature of the high-temperature hot bath is fixed as $\beta_h = 10$ and $\beta_h = 0.01$, respectively. In addition, the green solid line is the numerical reference line

5 Conclusions and Discussions

In conclusion, based on the quantum platform of a two-level system, we first introduce the assumption of low dissipation into the non-regenerative Stirling cycle and obtained many new results. Of course, in this paper, we also adopted some other assumptions and typical schemes. For the treatment of ideal isothermal processes, we use discrete isothermal processes (DIP) as an alternative, which has been widely adopted in previous research on quantum heat engines. We did not calculate the exact amount of heat exchanged in the isochoric process, but we compared them with the heat exchanged in the isothermal process and finally obtained the corresponding proportional coefficients ϕ and κ, respectively. Importantly, the general distribution of ϕ and κ is found in the new results obtained

First of all, we obtain the generalized efficiency boundary of the Stirling cycle under the assumption of low dissipation. We find that the coefficient $L = \frac{1+\phi}{1+\kappa}$ and the entropy change ratio α of the two isothermal processes affect the efficiency boundary range. In particular, when L is less than $-\alpha$, the efficiency of the heat engine at maximum power will exceed that of the upper bound $\frac{\eta_c}{2-\eta_c}$ of the low-dissipation Carnot cycle. Next, we find that the proportional coefficients corresponding to each isochoric process are almost asymmetrically equal($\phi \neq \kappa$). In the high-temperature region, $\phi < \kappa$, and in the low-temperature region, $\phi > \kappa$. This has guiding significance for the cyclic construction of heat engines. Finally, we study the variation of the heat engine efficiency in the case of symmetric dissipation($\frac{\Sigma_c}{\Sigma_h} = 1$) in the two regions. We find that in the small Carnot efficiency range, the efficiency at maximum power in the two regions is close to the CA efficiency, while in the large Carnot efficiency range, the heat engine efficiency in the high temperature region is the largest and the efficiency in the low temperature region is the smallest. In fact, the two-level heat engine we constructed can also be experimentally validated through other experimental platforms such as quantum dots [15, 16], ion traps [17], and superconducting circuit systems [49, 50]. Perhaps on different quantum platforms, our type of Stirling heat engine will have many new discoveries, which need to be verified in the future.

The authors confirm that the data supporting the findings of this study are available within the article.

Author Contributions The writing of this manuscript and the drawing of the pictures, as well as all other preparatory work, were completed by the first author (corresponding author) YangyangYuan alone.

Data Availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

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